

Worldsheet Conventions for Type IIB String Theory

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Contents

1	Worldsheet CFT conventions	2
1.1	Conventions for the free boson	2
1.2	Conventions for the free fermion	3
1.3	Conventions for the bc ghost system	5
1.4	Conventions for the $\beta\gamma$ system	6
1.5	Conventions for the TT and $T_F T_F$ operator products	8
1.6	Conventions for the BRST and PCO operators	9

1 Worldsheet CFT conventions

Euclidean worldsheet coordinates are denoted by z and $\bar{z}$. Target-space vector indices M, N take values in $\{0, 1, \dots, 9\}$. We use the ten-dimensional Minkowski metric

$$\eta_{MN} := \text{diag}(-1, +1, \dots, +1). \quad (1.1)$$

We set

$$\alpha' = 1. \quad (1.2)$$

Some other general CFT₂ conventions we adopt are as follows:

1. Every coincident-point product in these notes is understood to be normal ordered by point splitting. Thus the notation $A(w)B(w)$ means

$$A(w)B(w) := \lim_{z \rightarrow w} (A(z)B(w) - \text{Sing}_{z \rightarrow w}[A(z)B(w)]). \quad (1.3)$$

2. When a composite field is defined by an equation whose left-hand side is $C(z)$, every field on the right-hand side is evaluated at z unless a different argument is displayed explicitly.
3. If T is the stress tensor, a holomorphic operator $\mathcal{O}_h$ has conformal weight h when

$$T(z)\mathcal{O}_h(w) \sim \frac{h\mathcal{O}_h(w)}{(z-w)^2} + \frac{\partial\mathcal{O}_h(w)}{z-w}. \quad (1.4)$$

4. The central charge c of T is defined so that the coefficient of $(z-w)^{-4}$ in $T(z)T(w)$ is $c/2$.
5. A field of Grassmann parity $|A| \in \{0, 1\}$ is called even when $|A| = 0$ and odd when $|A| = 1$; away from operator-product singularities, two such fields obey $AB = (-1)^{|A||B|}BA$.

1.1 Conventions for the free boson

Let $X^M(z, \bar{z})$ be the ten embedding coordinates. Their two-point operator product and its differentiated holomorphic form are

$$X^M(z, \bar{z})X^N(w, \bar{w}) \sim -\frac{\eta^{MN}}{2} \log|z-w|^2, \quad (1.5)$$

$$\partial X^M(z)\partial X^N(w) \sim -\frac{\eta^{MN}}{2(z-w)^2}. \quad (1.6)$$

Equation (1.6) is the free-boson normalization used in Ref. [1]. The holomorphic stress tensor of the ten bosons is defined by

$$T_X(z) := -\eta_{MN}\partial X^M\partial X^N(z). \quad (1.7)$$

With Eqs. (1.6) and (1.7), each X^M has holomorphic conformal weight zero, each ∂X^M has holomorphic conformal weight one, and the ten-boson central charge is

$$c_X = 10. \quad (1.8)$$

For a target-space momentum $k_M \in \mathbb{R}^{1,9}$, define

$$k \cdot X := k_M X^M, \quad (1.9)$$

$$k^2 := \eta^{MN} k_M k_N, \quad (1.10)$$

$$V_k(z, \bar{z}) := \exp(\mathrm{i}k \cdot X(z, \bar{z})). \quad (1.11)$$

The holomorphic and anti-holomorphic conformal weights of V_k are

$$(h_k, \bar{h}_k) = \left(\frac{k^2}{4}, \frac{k^2}{4} \right). \quad (1.12)$$

Two useful consequences of Eq. (1.5) are

$$\partial X^M(z) V_k(w, \bar{w}) \sim -\frac{\mathrm{i}k^M}{2(z-w)} V_k(w, \bar{w}), \quad (1.13)$$

$$V_k(z, \bar{z}) V_p(w, \bar{w}) \sim |z-w|^{k \cdot p} \exp(\mathrm{i}(k+p) \cdot X(w, \bar{w})), \quad (1.14)$$

where $k \cdot p := \eta^{MN} k_M p_N$.

1.2 Conventions for the free fermion

Let $\psi^M(z)$ and $\tilde{\psi}^M(\bar{z})$ be the holomorphic and anti-holomorphic components of the ten real worldsheet fermions. Their normalization is

$$\psi^M(z) \psi^N(w) \sim -\frac{\eta^{MN}}{2(z-w)}. \quad (1.15)$$

The holomorphic fermion stress tensor is

$$T_\psi(z) := \eta_{MN} \psi^M \partial \psi^N(z). \quad (1.16)$$

Thus ψ^M has holomorphic conformal weight $1/2$, and the ten-fermion central charge is

$$c_\psi = 5. \quad (1.17)$$

The free matter stress tensor and the matter supercurrent are defined by

$$T_m := T_X + T_\psi, \quad (1.18)$$

$$T_F := -\eta_{MN}\psi^M\partial X^N. \quad (1.19)$$

We choose the convention in Eq. (1.19) to agree with Eq. (2.3) of Ref. [2].

We next fix the gamma matrices and the Ramond spin fields. Spinor indices α, β, γ take values in $\{1, \dots, 16\}$. We use symmetric chiral gamma-matrix blocks

$$(\Gamma^M)_{\alpha\beta} = (\Gamma^M)_{\beta\alpha}, \quad (\Gamma^M)^{\alpha\beta} = (\Gamma^M)^{\beta\alpha}. \quad (1.20)$$

Their normalization is defined by

$$(\Gamma^M)_{\alpha\beta}(\Gamma^N)^{\beta\gamma} + (\Gamma^N)_{\alpha\beta}(\Gamma^M)^{\beta\gamma} = 2\eta^{MN}\delta_\alpha^\gamma. \quad (1.21)$$

Equivalently, the 32×32 matrices

$$\mathbf{\Gamma}^M := \begin{pmatrix} 0 & (\Gamma^M)^{\alpha\beta} \\ (\Gamma^M)_{\alpha\beta} & 0 \end{pmatrix} \quad (1.22)$$

satisfy

$$\{\mathbf{\Gamma}^M, \mathbf{\Gamma}^N\} = 2\eta^{MN}\mathbf{1}_{32}. \quad (1.23)$$

For $p \geq 1$, the antisymmetrized product is

$$\mathbf{\Gamma}^{M_1 \dots M_p} := \mathbf{\Gamma}^{[M_1} \dots \mathbf{\Gamma}^{M_p]}, \quad (1.24)$$

where the brackets include the factor $1/p!$. A symbol such as $(\Gamma^{M_1 \dots M_p})_\alpha^\beta$ denotes the indicated chiral block of Eq. (1.24). We choose the target-space orientation and positive-chirality block by

$$\epsilon^{01 \dots 9} = +1, \quad (\Gamma^{01 \dots 9})_\alpha^\beta = \delta_\alpha^\beta. \quad (1.25)$$

Equations (1.20) and (1.25) are the ten-dimensional conventions of Ref. [3].

Let $S_\alpha(z)$ and $\tilde{S}_\alpha(\bar{z})$ denote the left- and right-moving positive-chirality Ramond spin fields. Type IIB means that these two fields have the same ten-dimensional chirality. Each has conformal weight $5/8$ in its chiral sector. The symbols S^α and $\tilde{S}^\alpha$ denote the spin fields that occur in the $-3/2$ picture; the placement of the spinor index is part of this convention and is

fixed by the operator products below, rather than by an unstated index-raising rule.

The Ramond operator products also contain the bosonized superghost ϕ . The scalar ϕ is Grassmann even. The symbol $e^{q\phi}$ denotes its exponential together with the cocycle factor, namely the z -independent factor that supplies the Grassmann parity required by the bosonization. For integer q , we define

$$|e^{q\phi}| := q \pmod{2}, \quad q \in \mathbb{Z}.$$

Thus $e^{q\phi}$ is Grassmann odd when q is odd. For half-integer q , a Grassmann parity is assigned only to the complete Ramond operator containing $e^{q\phi}$ and a spin field, rather than to $e^{q\phi}$ by itself. For $q_1, q_2 \in \frac{1}{2}\mathbb{Z}$, the scalar and exponential operator products used below are

$$\begin{aligned} \partial\phi(z)\partial\phi(w) &\sim -\frac{1}{(z-w)^2}, \\ e^{q_1\phi}(z)e^{q_2\phi}(w) &\sim (z-w)^{-q_1q_2}e^{(q_1+q_2)\phi}(w). \end{aligned} \tag{1.26}$$

The complete $\beta\gamma$ bosonization is given in Sec. 1.4. We normalize the spin fields by the singular terms

$$e^{-\phi}\psi^M(z)e^{-\phi/2}S_\alpha(w) \sim \frac{i}{2(z-w)}(\Gamma^M)_{\alpha\beta}e^{-3\phi/2}S^\beta(w), \tag{1.27}$$

$$e^{-3\phi/2}S^\alpha(z)e^{-\phi/2}S_\beta(w) \sim \frac{\delta^\alpha_\beta}{(z-w)^2}e^{-2\phi}(w), \tag{1.28}$$

$$e^{-\phi/2}S_\alpha(z)e^{-\phi/2}S_\beta(w) \sim \frac{i}{z-w}(\Gamma^M)_{\alpha\beta}e^{-\phi}\psi_M(w). \tag{1.29}$$

The branch choices, phases, and factors in Eqs. (1.27)–(1.29) are those of Ref. [3]. The anti-holomorphic spin fields obey the same equations with every field and coordinate replaced by its anti-holomorphic counterpart. This identical left-right form is another place where the type IIB chirality choice enters.

1.3 Conventions for the bc ghost system

The reparametrization ghosts $b(z)$ and $c(z)$ are Grassmann odd. Their holomorphic conformal weights and ghost numbers are

field	h	ghost number
b	2	−1
c	−1	+1

Their singular operator product is

$$b(z)c(w) \sim \frac{1}{z-w}. \quad (1.30)$$

The bc stress tensor is defined by

$$T_{bc} := -2b\partial c - (\partial b)c, \quad (1.31)$$

and its central charge is

$$c_{bc} = -26. \quad (1.32)$$

On the complex plane, a nonzero bc correlator contains three more c insertions than b insertions. We define the constant K by the normalization used in Ref. [1]:

$$\langle c(z_1)c(z_2)c(z_3) \rangle_{bc} := -K(z_1 - z_2)(z_2 - z_3)(z_1 - z_3). \quad (1.33)$$

Here the subscript bc means that the expectation value is taken only in the bc sector. Equation (1.33) defines K ; its relation to a D-brane or D-instanton tension additionally depends on the normalization of the string amplitude and is not assumed in these notes.

1.4 Conventions for the $\beta\gamma$ system

The superconformal ghosts $\beta(z)$ and $\gamma(z)$ are Grassmann even and have conformal weights $3/2$ and $-1/2$, respectively. We bosonize them in terms of Grassmann-odd fields ξ, η and the Grassmann-even chiral scalar ϕ by

$$\beta := \partial\xi e^{-\phi}, \quad (1.34)$$

$$\gamma := \eta e^{\phi}. \quad (1.35)$$

The cocycle convention preceding Eq. (1.26) makes both $e^{-\phi}$ and e^{ϕ} Grassmann odd. Consequently, each right-hand side in Eqs. (1.34) and (1.35) is Grassmann even. The displayed factor order is part of the convention. The elementary operator products are

$$\xi(z)\eta(w) \sim \frac{1}{z-w}, \quad (1.36)$$

$$\partial\phi(z)\partial\phi(w) \sim -\frac{1}{(z-w)^2}, \quad (1.37)$$

$$e^{q_1\phi}(z)e^{q_2\phi}(w) \sim (z-w)^{-q_1q_2}e^{(q_1+q_2)\phi}(w). \quad (1.38)$$

Graded locality and differentiation of Eq. (1.36) give

$$\begin{aligned}\partial\xi(z)\eta(w) &\sim -\frac{1}{(z-w)^2}, \\ \eta(z)\partial\xi(w) &\sim \frac{1}{(z-w)^2}.\end{aligned}$$

Moving the odd exponential through the odd η or $\partial\xi$ therefore gives

$$\begin{aligned}\beta(z)\gamma(w) &= \partial\xi(z)e^{-\phi}(z)\eta(w)e^{\phi}(w) \\ &= -\partial\xi(z)\eta(w)e^{-\phi}(z)e^{\phi}(w) \sim \frac{1}{z-w}, \\ \gamma(z)\beta(w) &= \eta(z)e^{\phi}(z)\partial\xi(w)e^{-\phi}(w) \\ &= -\eta(z)\partial\xi(w)e^{\phi}(z)e^{-\phi}(w) \sim -\frac{1}{z-w}.\end{aligned}$$

Thus

$$\begin{aligned}\beta(z)\gamma(w) &\sim \frac{1}{z-w}, \\ \gamma(z)\beta(w) &\sim -\frac{1}{z-w}.\end{aligned}\tag{1.39}$$

The bosonized stress tensors are

$$T_{\xi\eta} := -\eta\partial\xi,\tag{1.40}$$

$$T_{\phi} := -\frac{1}{2}(\partial\phi)^2 - \partial^2\phi,\tag{1.41}$$

$$T_{\beta\gamma} := T_{\xi\eta} + T_{\phi}.\tag{1.42}$$

In the unbosonized variables, the same stress tensor is

$$T_{\beta\gamma} = \frac{3}{2}\beta\partial\gamma + \frac{1}{2}\gamma\partial\beta.\tag{1.43}$$

Its central charge is

$$c_{\beta\gamma} = 11.\tag{1.44}$$

The ghost number and picture number used below are additive charges defined on the elementary bosonized fields by the following table.

field	β	γ	ξ	η	$e^{q\phi}$
ghost number	-1	+1	-1	+1	0
picture number	0	0	+1	-1	q
Grassmann parity	0	0	1	1	$q \bmod 2$

The last entry in the Grassmann-parity row applies to $q \in \mathbb{Z}$. The conformal weight of the exponential, for $q \in \frac{1}{2}\mathbb{Z}$, is

$$h[e^{q\phi}] = -\frac{1}{2}q(q+2). \quad (1.45)$$

The mode expansions of the weight-zero field ξ and the weight-one field η define

$$\xi_0 := \oint \frac{dz}{2\pi i z} \xi(z), \quad \eta_0 := \oint \frac{dz}{2\pi i} \eta(z). \quad (1.46)$$

The large Hilbert space is the state space on which ξ_0 acts without the restriction in Eq. (1.47). The small Hilbert space is the subspace of states $|\Psi\rangle$ satisfying

$$\eta_0|\Psi\rangle = 0. \quad (1.47)$$

Equations (1.34)–(1.47) follow the ordering, charges, and Hilbert-space convention of Ref. [1]. The cocycle grading and the positive coefficients in Eq. (1.43) agree with Eqs. (3.17)–(3.20) of Ref. [4]; the parity discussion after Eq. (3.27) of that reference assigns $e^{q\phi}$ the corresponding $(-1)^q$ GSO parity for integer q .

1.5 Conventions for the TT and $T_F T_F$ operator products

Let $T(z)$ be the stress tensor and $T_F(z)$ be its supercurrent in the normalization defined below, and let c be its central charge. We define the normalization of T and T_F by the singular operator products

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \quad (1.48)$$

$$T(z)T_F(w) \sim \frac{3}{2} \frac{T_F(w)}{(z-w)^2} + \frac{\partial T_F(w)}{z-w}, \quad (1.49)$$

$$T_F(z)T_F(w) \sim \frac{c/6}{(z-w)^3} + \frac{T(w)}{2(z-w)}. \quad (1.50)$$

Equations (1.48)–(1.50) match the convention used in Ref. [1].

1.6 Conventions for the BRST and PCO operators

The holomorphic BRST current is defined by

$$j_B := c(T_m + T_{\xi\eta} + T_\phi) + bc\partial c + \gamma T_F - \frac{1}{4}\gamma^2 b. \quad (1.51)$$

In bosonized variables, the composite field γ^2 is

$$\gamma^2 = \eta \partial \eta e^{2\phi}. \quad (1.52)$$

The BRST charge is the contour integral

$$Q_B := \oint \frac{dz}{2\pi i} j_B(z), \quad (1.53)$$

where the contour is counterclockwise around the operator on which Q_B acts. The anti-holomorphic charge $\tilde{Q}_B$ is defined by the tilded copy of Eqs. (1.51)–(1.53); the closed-string BRST charge is

$$Q_{\text{closed}} := Q_B + \tilde{Q}_B. \quad (1.54)$$

For two Grassmann-odd operators A and B , write $\{A, B\} := AB + BA$. The holomorphic picture-changing operator (PCO) is defined in the large Hilbert space by

$$\mathcal{X}(z) := 2\{Q_B, \xi(z)\} \quad (1.55)$$

$$= 2c\partial\xi + 2e^\phi T_F - \frac{1}{2}[(\partial\eta)e^{2\phi}b + \partial(\eta e^{2\phi}b)](z). \quad (1.56)$$

The factor of two in Eq. (1.55) is part of the convention adopted from Ref. [1]. The PCO $\mathcal{X}$ has conformal weight zero, ghost number zero, picture number +1, and is BRST closed. The anti-holomorphic PCO is

$$\tilde{\mathcal{X}}(\bar{z}) := 2\{\tilde{Q}_B, \tilde{\xi}(\bar{z})\}. \quad (1.57)$$

For an oriented worldsheet of genus $g \in \mathbb{Z}_{\geq 0}$ with $b \in \mathbb{Z}_{\geq 0}$ boundary components, define P_{tot} to be the sum of the picture numbers of all holomorphic and anti-holomorphic insertions, including PCOs. A nonzero correlator requires

$$P_{\text{tot}} = 2(2g + b - 2). \quad (1.58)$$

For a closed worldsheet, $b = 0$, this condition can be imposed separately as $P_L = P_R = 2g - 2$. Equation (1.58) explains why PCO insertions are required when the picture numbers of the vertex operators do not already have the required sum.

References

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